

Elementary part of Friction (Tutorial Prob)

Exp- 4.2 p234

Exp- 4.3, p-235

Exp- 4.5, p-236-237

Exp- 4.9, p-240-241

Exp- 4.13, p-244-245

Exp- 4.35, p-265

Adding Eqs (2) and (4), we obtain, $(N_1 + N_2) - (W_1 + W_2) + P \sin \alpha = 0$. Substituting in Eq. (5),

$$P = \frac{\mu(W_1 + W_2)}{(\cos \alpha + \mu \sin \alpha)} = 0 \quad (6)$$

If P is to be minimum, $\frac{dP}{d\alpha} = 0$, for which $(\cos \alpha + \mu \sin \alpha)$ will be maximum

So, $\frac{d}{d\alpha} (\cos \alpha + \mu \sin \alpha) = 0$ or $\mu = \tan \alpha$

As per given condition, $\mu = \tan \phi$

Therefore, $\alpha = \phi$

$$\text{Hence, } P_{\min} = \frac{\tan \phi (W_1 + W_2)}{(\cos \phi + \tan \phi \sin \phi)} = (W_1 + W_2) \sin \phi$$

Example 4.2 Two blocks connected with a tie rod are shown in Fig. E4.2(a). If $\phi_B = 15^\circ$ and $\mu_A = 0.40$, find the smallest value of W for the equilibrium of the system.

Solution The compressive force acting on bar is assumed as T .

Consider FBD of block B [Fig. E4.2(b)]

From the equilibrium of vertical forces:

$$\uparrow \sum F_y = 0: \quad F_B \cos 30^\circ + N_B \sin 30^\circ - 400 = 0$$

$$\text{or } \mu_B N_B \cos 30^\circ + N_B \sin 30^\circ - 400 = 0$$

$$\text{or } \tan \phi_B N_B \cos 30^\circ + N_B \sin 30^\circ - 400 = 0$$

$$\text{or } N_B = 546.41 \text{ N}$$

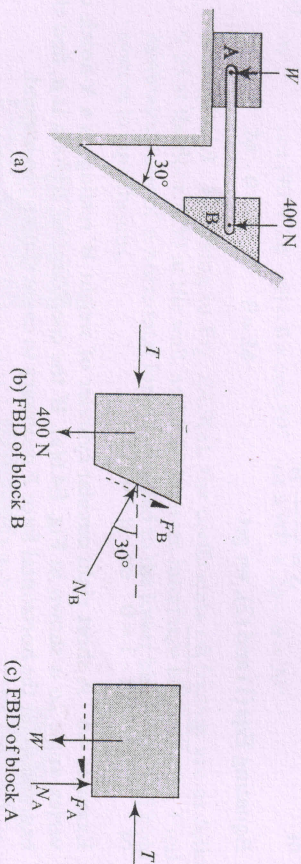


Fig. E4.2

From the equilibrium of horizontal forces:

$$\sum F_x = 0: \quad T - N_B \cos 30^\circ + F_B \sin 30^\circ = 0$$

$$\text{or } T - N_B \cos 30^\circ + \tan \phi_B N_B \sin 30^\circ = 0$$

$$\text{or } T = 400 \text{ N}$$

Consider FBD of block A [Fig. E4.2(c)]

From the equilibrium of horizontal forces:

$$\sum F_x = 0: \quad T = F_A = \mu_A N_A = 0.4 N_A$$

From the equilibrium of vertical forces:

$$\uparrow \sum F_y = 0: \quad N_A = W$$

On substitution, $T = 0.4 W$

$$\text{So, } W = \frac{400}{0.4} = 1000 \text{ N}$$

Example 4.3 Two blocks of weights W_1 and W_2 connected with a string rest on a rough incline, as shown in Fig. E4.3(a). If the coefficients of friction are 0.2 and 0.3 for the blocks, respectively, and $W_1 = W_2 = 50 \text{ N}$, find the value of α for which sliding will impend.

Solution Assume the tension of the string to be S . Equilibrium of the blocks will be considered along the incline and perpendicular to the incline.

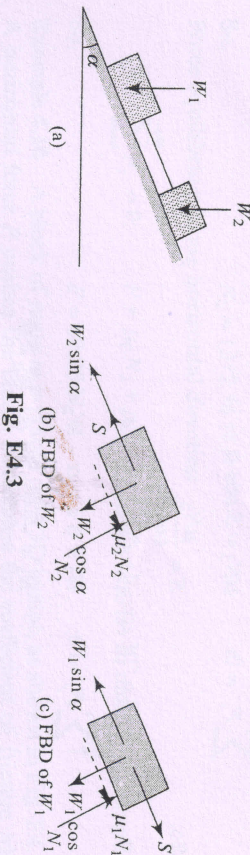


Fig. E4.3

Consider FBD of W_2 [Fig. E4.3(b)]

Equilibrium perpendicular to the incline:

$$N_2 = W_2 \cos \alpha$$

Equilibrium along the incline:

$$-S - W_2 \sin \alpha + \mu_2 N_2 = 0$$

$$\text{So, } S = \mu_2 W_2 \cos \alpha - W_2 \sin \alpha$$

Consider FBD of W_1 [Fig. E4.3(c)]

Equilibrium perpendicular to the incline:

$$N_1 = W_1 \cos \alpha$$

Equilibrium along the incline:

$$S + \mu_1 N_1 = W_1 \sin \alpha$$

$$\text{So, } S = W_1 \sin \alpha - \mu_1 W_1 \cos \alpha$$

Equating Eqs (1) and (2):

$$\begin{aligned} \mu_2 W_2 \cos \alpha - W_2 \sin \alpha &= W_1 \sin \alpha - \mu_1 W_1 \cos \alpha \\ (W_1 + W_2) \sin \alpha &= (\mu_1 W_1 + \mu_2 W_2) \cos \alpha \\ \tan \alpha &= \frac{\mu_1 W_1 + \mu_2 W_2}{W_1 + W_2} = \frac{(0.2 + 0.3) 50}{2 \times 50} = 0.25 \end{aligned}$$

Example 4.4 A block of weight $W_1 = 200 \text{ N}$ rests on a horizontal surface and supports on top of it another block of weight $W_2 = 50 \text{ N}$. The block W_2 attached with vertical wall by a string AB, is shown in Fig. E4.4(a). Find the amount of horizontal force P , applied to the lower block necessary for impending slipping. The coefficient of friction for all contiguous surfaces is 0.3.

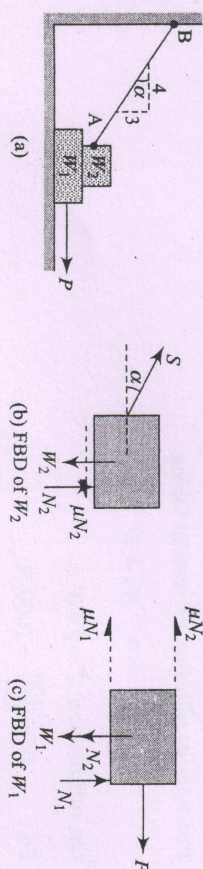


Fig. E4.4

Solution Here, $\tan \alpha = 3/4$. So, $\sin \alpha = 3/5$ and $\cos \alpha = 4/5$. Assume that the tension in the string is S .

Consider FBD of W_2 [Fig. E4.4(b)]

Force equilibrium in horizontal direction:

$$\sum \vec{F}_x = 0: \quad \mu N_2 = S \cos \alpha$$

So,

$$S = \frac{\mu N_2}{\cos \alpha}$$

Force equilibrium in vertical direction:

$$\sum \overset{\uparrow}{F}_y = 0: \quad N_2 + S \sin \alpha - W_2 = 0$$

$$N_2 + \frac{\mu N_2}{\cos \alpha} \sin \alpha - W_2 = 0$$

Substituting the values, we obtain, $N_2 = 40.816$ N.

Consider FBD of W_1 [Fig. E4.4(c)]

Force equilibrium in vertical direction:

$$\sum \overset{\uparrow}{F}_y = 0: \quad N_1 = N_2 + W_1 = 240.816 \text{ N}$$

Force equilibrium along horizontal direction:

$$\sum \vec{F}_x = 0: \quad P = \mu N_1 + \mu N_2 = 0.3(40.816 + 240.816) = 84.49 \text{ N}$$

Example 4.5 Two identical blocks, A and B, are on the vertical wall and B on the floor are kept stationary with the help of a connected strut, as shown in Fig. E4.5(a). If sliding impends when $\theta = 45^\circ$, determine the coefficient of friction, assuming it to be same at both the floor and the wall.

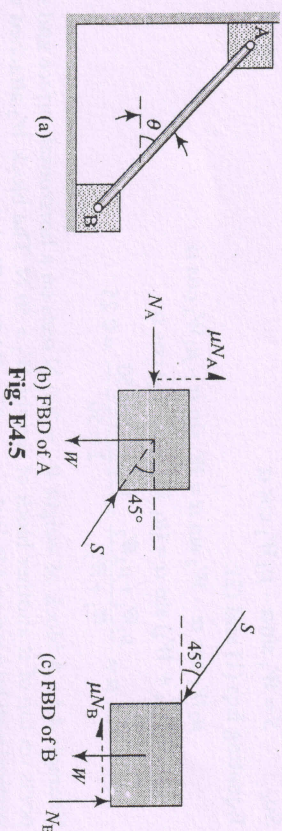


Fig. E4.5

Solution Let the compressive force in the strut be S , and the weight of each block be W .

Consider FBD of block A [Fig. E4.5(b)]

Force equilibrium along horizontal direction:

$$\sum \vec{F}_x = 0: \quad N_A = S \cos 45^\circ$$

Force equilibrium along vertical direction:

$$\sum \overset{\uparrow}{F}_y = 0: \quad S \sin 45^\circ + \mu N_A - W = 0$$

$$S(\sin 45^\circ + \mu \cos 45^\circ) - W = 0$$

So,

$$S = \frac{\sqrt{2}}{1 + \mu} W \quad (1)$$

Consider FBD of block B [Fig. E4.5(c)]

Force equilibrium along horizontal direction:

$$\sum \vec{F}_x = 0: \quad S \cos 45^\circ - \mu N_B = 0$$

So,

$$N_B = \frac{S}{\mu \sqrt{2}}$$

Force equilibrium along vertical direction:

$$\sum \overset{\uparrow}{F}_y = 0: \quad N_B - S \sin 45^\circ - W = 0$$

or

$$\frac{S}{\mu \sqrt{2}} - \frac{S}{\sqrt{2}} - W = 0$$

or

$$S = \frac{\mu \sqrt{2}}{1 - \mu} W$$

(2)

Equating Eqs (1) and (2), we get

$$\frac{1}{1 + \mu} = \frac{\mu}{1 - \mu}$$

$$\text{or } \mu^2 + 2\mu - 1 = 0 \quad \text{or } \mu = 0.414$$

Example 4.6 A short right circular cylinder of weight W resting on a V-notch of vertex angle 2α is shown in Fig. E4.6(a). If the coefficient of friction is μ , find the expression for the horizontal force P necessary to cause slipping to impend.

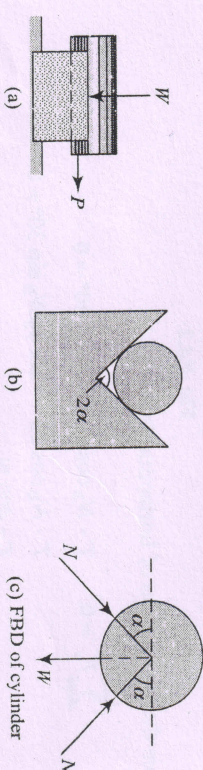


Fig. E4.6

Solution Side view of the cylinder on notch is shown in Fig. E4.6(b). We can say from symmetry that the reaction coming from notch surface on the cylinder on either side will be equal.

Consider FBD of cylinder [Fig. E4.6(c)]

$$N_B \cos 45^\circ - \mu N_B \cos 45^\circ \sin 45^\circ - N_A \cos 45^\circ - \mu N_A \cos 45^\circ \sin 45^\circ = 0$$

Being C, mid-point of AB, $AC = CB$

$$\text{So, } (N_A - N_B) + \mu(N_A + N_B) = 0$$

Substituting Eq. (4) in Eq. (3), we obtain

$$\tan \theta = \frac{2\mu}{1 - \mu^2} = \frac{2 \tan \phi}{1 - \tan^2 \phi} = \tan 2\phi$$

$$\text{So, } \theta = 2\phi$$

Geometrical approach

For equilibrium, the resultant reactions at A and B, respectively, i.e., R_A , R_B , and self-weight of the bar should be concurrent. Let us assume the length of bar AB is l . Now, $\angle CAB = (45^\circ + \phi)$, $\angle CBA = (45^\circ - \phi)$

$$\angle CAD = (45^\circ + \phi - \theta), \angle ACB = 90^\circ, AB = l, AG = l/2$$

From ADAG,

$$AD = \frac{l}{2} \cos \theta \quad (1)$$

From $\triangle CAB$,

$$AC = l \cos(45^\circ + \phi)$$

Again from $\triangle CAD$,

$$AD = AC \cos \angle CAD = l \cos(45^\circ + \phi) \cos(45^\circ + \phi - \theta) \quad (2)$$

Equating Eqs (1) and (2), we obtain

$$\frac{l}{2} \cos \theta = l \cos(45^\circ + \phi) \cos(45^\circ + \phi - \theta)$$

$$\text{or } \cos \theta = 2 \cos(45^\circ + \phi) \cos(45^\circ + \phi - \theta)$$

$$= \cos \theta + \cos(90^\circ + 2\phi - \theta)$$

$$\text{or } \cos(90^\circ + 2\phi - \theta) = 0 = \cos 90^\circ \quad \text{or } 90^\circ + 2\phi - \theta = 90^\circ$$

$$\text{or } 2\phi - \theta = 0 \quad \text{or } \theta = 2\phi$$

Example 4.9 Referring to Fig. E4.9(a), the coefficients of friction are as follows: 0.25 at the floor, 0.30 at the wall, and 0.20 between interfaces of the blocks. Find the minimum value of a horizontal force P applied to the lower block that will hold the system in equilibrium.

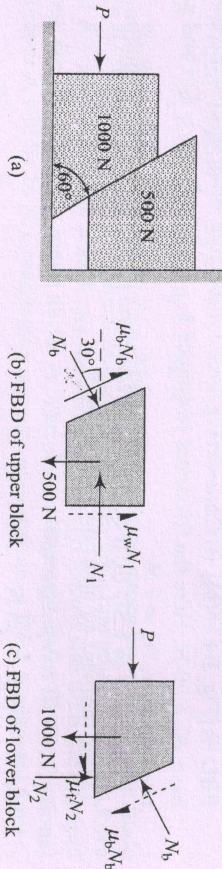


Fig. E4.9

Solution Coefficient of friction of floor $\mu_f = 0.25$, at wall $\mu_w = 0.3$, and between blocks $\mu_b = 0.2$.

Consider FBD of upper block [Fig. E4.9(b)]

Force equilibrium along horizontal direction:

$$\sum F_x = 0: \quad N_b \cos 30^\circ - \mu_b N_b \cos 60^\circ - N_1 = 0$$

$$\text{So, } N_1 = 0.766 N_b$$

Force equilibrium along vertical direction:

$$\uparrow \sum F_y = 0: \quad \mu_b N_b \sin 60^\circ + \mu_w N_1 + N_b \sin 30^\circ - 500 = 0$$

Substituting N_1 , we have:

$$\mu_b N_b \sin 60^\circ + \mu_w \times 0.766 N_b + N_b \sin 30^\circ - 500 = 0$$

$$\text{So, } N_b = 553.706 \text{ N}$$

Consider FBD of lower block [Fig. E4.9(c)]

Force equilibrium along vertical direction:

$$\uparrow \sum F_y = 0: \quad N_2 - N_b \sin 30^\circ - \mu_b N_b \sin 60^\circ - 1000 = 0$$

$$\text{So, } N_2 = 1372.76 \text{ N}$$

Force equilibrium along horizontal direction:

$$\sum F_x = 0: \quad P + \mu_f N_2 + \mu_b N_b \cos 60^\circ - N_b \cos 30^\circ = 0$$

$$\text{So, } P = 80.96 \text{ N}$$

Example 4.10 A block of mass m is placed on a 3D incline, as shown in Fig. E4.10. A horizontal force P pushes the block. Determine the coefficient of friction when $P = 0.4 mg$.

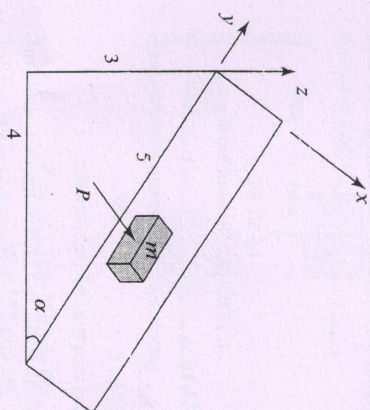


Fig. E4.10

Solution As P pushes the block along x -direction, frictional resistance, say F_x , will act along x -direction, opposite to P . So, $F_x = P = 0.4 mg$

Again, the block has a natural tendency to slide down, thus generating a frictional resistance, say F_y , along y -direction, opposite to the direction of in-plane component of the self-weight of block.

$$\text{Thus, } F_y = mg \sin \alpha = \frac{3}{5} mg = 0.6 mg.$$

Now, considering force equilibrium along z -direction

$$N = mg \cos \alpha = \frac{4}{5} mg = 0.8 mg$$

Hence, the net frictional force will be the

resultant of F_x and F_y .

$$\text{So, } F = \sqrt{F_x^2 + F_y^2}$$

$$= \sqrt{(0.4mg)^2 + (0.6mg)^2}$$

$$\text{or } \mu N = 0.7211 mg$$

$$\text{or } \mu = \frac{0.7211 mg}{N} = \frac{0.7211 mg}{0.8 mg} = 0.9013$$

$$150 \sin 45^\circ + 0.2679 N_A = F \cos 15^\circ$$

$$150 \cos 45^\circ + F \sin 15^\circ = N_A$$

Solving the above equations, we obtain $F = 149.96 \text{ N}$.

Consider FBD of block B [Fig. E4.12(c)]

Take force equilibrium along horizontal and vertical directions:

$$\sum \vec{F}_x = 0: \quad F \cos 30^\circ + \mu N_B = P$$

$$\sum \vec{F}_y = 0: \quad F \sin 30^\circ + 450 = N_B$$

So,

$$\sum \vec{F}_y = 0: \quad F \sin 30^\circ + 450 = N_B$$

Solving the above two equations, we obtain $P = 270.5 \text{ N}$.

Example 4.13 Referring to Fig. E4.13(a), determine the least value of P to cause motion to impend rightwards. Assume the pulley to be frictionless and coefficient of friction of all contiguous surfaces is 0.20.

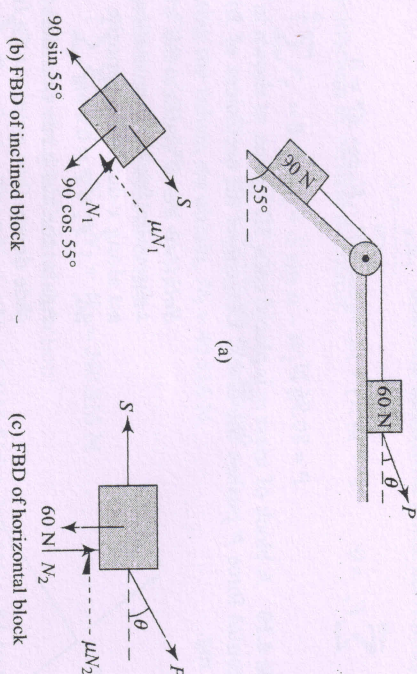


Fig. E4.13

Solution Let the tensile force in the string be S .

Consider FBD of inclined block [Fig. E4.13(b)]

Take force equilibrium along the incline and normal to the incline:

$$90 \sin 55^\circ + \mu N_1 = S \quad \text{or} \quad 90 \sin 55^\circ + 0.2 N_1 = S$$

$$90 \cos 55^\circ = N_1 \quad 90 \cos 55^\circ = N_1$$

Solving these equations, we get $N_1 = 51.62 \text{ N}$ and $S = 84.05 \text{ N}$.

Consider FBD of horizontal block [Fig. E4.13(c)]

Take force equilibrium along horizontal and vertical directions:

$$\sum \vec{F}_x = 0: \quad S + \mu N_2 = P \cos \theta$$

$$\sum \vec{F}_y = 0: \quad 84.05 + 0.2 N_2 = P \cos \theta$$

$$\sum \vec{F}_y = 0: \quad N_2 + P \sin \theta = 60$$

Solving the above equations, we get

$$P = \frac{96.05}{(\cos \theta + 0.2 \sin \theta)}$$

For the least value of P , the magnitude of $(\cos \theta + 0.2 \sin \theta)$ should be maximum.

$$\text{Hence } \frac{d}{d\theta} (\cos \theta + 0.2 \sin \theta) = 0 \quad \text{or} \quad -\sin \theta + 0.2 \cos \theta = 0$$

$$\text{or } \tan \theta = 0.2 \quad \text{or } \theta = 11.309^\circ$$

$$\text{Therefore, } P_{\min} = \frac{96.05}{(\cos 11.309^\circ + 0.2 \sin 11.309^\circ)} = 94.185 \text{ N}$$

Example 4.14 A uniform ladder of length 10 m weighing 20 N is placed against a smooth vertical wall with its lower end 4 m away from the wall [Fig. E4.14(a)]. The coefficient of friction between the ladder and the floor is 0.3. Check whether or not the ladder will remain in equilibrium in this position.

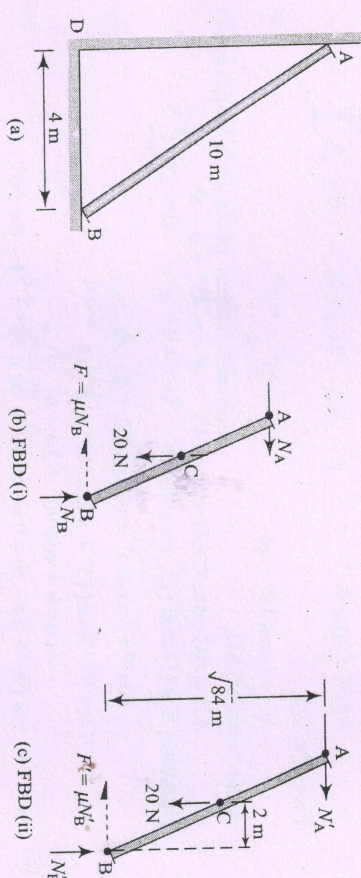


Fig. E4.14

Solution From right-angled triangle ADB, $AD = \sqrt{10^2 - 4^2} = \sqrt{84} \text{ m}$. The weight of the ladder will act at mid-length of AB, i.e., at point C. If the ladder just deviates from static equilibrium, it will have a tendency to slide down. This case is shown in FBD (i) [Fig. E4.14(b)]. (i) Considering force equilibrium along vertical direction, we have, $N_B = 20 \text{ N}$. Here the frictional resistance $F = \mu N_B = 0.3 \times 20 = 6 \text{ N}$ and normal reaction at A, $N_A = 6 \text{ N}$.

Now refer to FBD (ii) [Fig. E4.14(c)], where the ladder remains in equilibrium.

Taking moment equilibrium at B, we have

$$20 \times 2 = N_A \times \sqrt{84}$$

$$\text{or } N_A = 4.36 \text{ N}$$

$$\text{or } F' = \mu N_B = N_A = 4.36 \text{ N}$$

This frictional resistance is below F , which is the frictional resistance at impending motion. Hence, the ladder will remain in equilibrium in the given position.

Example 4.15 A ladder 8 m long weighing 200 N is resting against a rough vertical wall as shown in Fig. E4.15(a). A man of 720 N climbs the ladder at M. At what position will slipping be induced? Take μ for all surfaces 0.25.

$$\sum M_{at A} = 0: R_B \times x = 300 \times 4$$

$$\text{or } R_B = \frac{1200}{x} \quad (1)$$

$$\text{So } R_A = 300 - \frac{1200}{x} \quad (2)$$

For $x = 6$ m, $R_B = 200$ N and $R_A = 100$ N.

The plank may rotate about either A or B. If it rotates about A, the frictional force at B will be limiting. So, limiting frictional force at B

$$F_B = \mu \times R_B = 0.4 \times 200 = 80 \text{ N}$$

Consider moment equilibrium at A for horizontal force

$$\sum M_{at A} = 0: F_B \times 6 = P \times 8$$

$$\text{or } P = \frac{80 \times 6}{8} = 60 \text{ N}$$

For equilibrium, $F_A + F_B = P$

$$\text{or } F_A = P - F_B = 60 - 80 = -20 \text{ N}$$

So, the correct direction of F_A will be reverse to the assumed one and of magnitude 20 N. This is the required F_A for equilibrium. But the available maximum friction force at A = $\mu R_A = 0.4 \times 100 = 40$ N. As the available force is greater than the required one, it can be inferred that the plank will not rotate about A and the required amount of P is 60 N.

For the second case, when $x = 5$ m

$$R_B = 240 \text{ N and } R_A = 60 \text{ N}$$

[Substituting from Eqs (1) and (2)]

If the plank rotates about A, limiting friction force at B

$$F_B = \mu \times R_B = 0.4 \times 240 = 96 \text{ N}$$

Consider moment equilibrium at A for horizontal force

$$\sum M_{at A} = 0: F_B \times 5 = P \times 8$$

$$\text{or } P = \frac{96 \times 5}{8} = 60 \text{ N}$$

For equilibrium, $F_A + F_B = P$

$$\text{or } F_A = P - F_B = 60 - 96 = -36 \text{ N}$$

So, the correct direction of F_A will be reverse to the assumed one and of magnitude 36 N. This is the required F_A for equilibrium. The available maximum frictional force at A = $\mu R_A = 0.4 \times 60 = 24$ N. As the available friction force is less than the required one, the plank will not rotate about A. The plank will rotate about B. The frictional force at A will be 24 N. Considering moment equilibrium at B for horizontal force

$$\sum M_{at B} = 0: 24 \times 5 = P \times (8 - 5)$$

$$\text{or } P = \frac{24 \times 5}{3} = 40$$

Hence magnitude of required P is 40 N.

(ii) To determine $x_{\min}$, proceeding as before, we obtain

$$R_B = \frac{1200}{x_{\min}} \text{ and } R_A = 300 - \frac{1200}{x_{\min}}$$

The given condition states that the plank will slip at B. So, frictional force at B

$$F_B = \mu R_B = 0.4 \times \frac{1200}{x_{\min}} = \frac{480}{x_{\min}}$$

Consider moment equilibrium at A for horizontal force

$$F_B \times x_{\min} = P \times 8$$

$$\text{or } P = \frac{480}{x_{\min}} \times \frac{1}{8} = 60 \text{ N}$$

$$\text{So, frictional force at A, } F_A = F_B - P = \left(\frac{480}{x_{\min}} - 60 \right) \text{ N}$$

The limiting frictional force at A

$$= 0.4 \times R_A = 0.4 \left(300 - \frac{1200}{x_{\min}} \right) = \left(120 - \frac{480}{x_{\min}} \right) \text{ N}$$

To obtain $x_{\min}$ for which the plank slips at B, we need to equate the required frictional force at A with the limiting frictional force at A.

$$\text{So, } \frac{480}{x_{\min}} - 60 = 120 - \frac{480}{x_{\min}}$$

$$\text{or } \frac{960}{x_{\min}} = 180 \text{ or } x_{\min} = 5.334$$

So, the minimum value of x is 5.334 m.

Example 4.35 Block A weighing 240 kg resting on a rough floor supports block B weighing 120 kg as shown in Fig. E4.35(a). Both the blocks are connected with a rope passing over a smooth pulley. Compute the magnitude of force P at impending motion and the tension induced in the rope, if the coefficient of friction for all contiguous surfaces is 0.3.

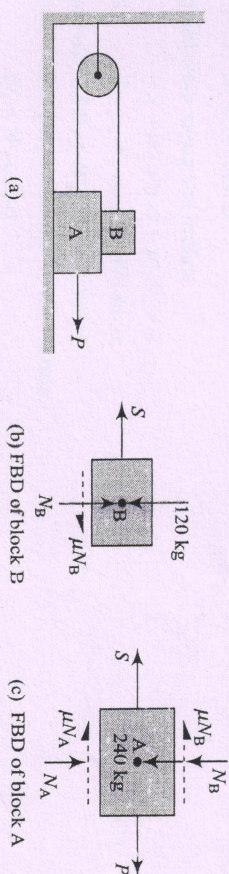


Fig. E4.35

Solution Consider FBD of block B [Fig. E4.35(b)]

Taking the condition of vertical force of equilibrium $N_B = 120$ kg

From the condition of horizontal force of equilibrium,

$$S = \mu N_B = 0.3 \times 120 = 36 \text{ kg}$$

Consider FBD of block A [Fig. E4.35(c)]

Take condition of force equilibrium along the vertical:

$$N_A = N_B + 240 = 120 + 240 = 360 \text{ kg}$$

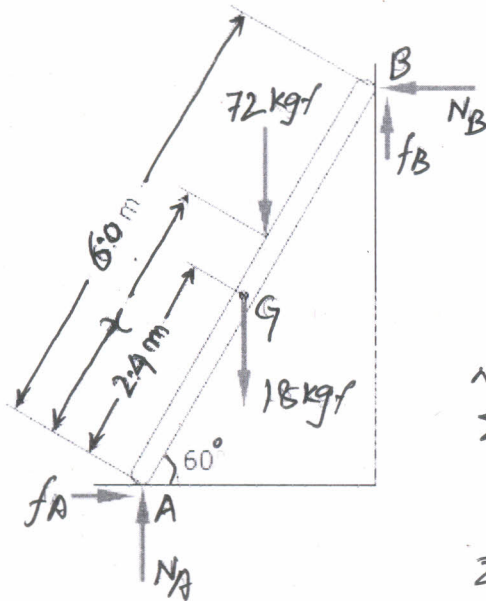
Again, from the condition of force equilibrium along the horizontal,

$$P = \mu N_A + \mu N_B + S$$

$$= 0.3 \times 360 + 0.3 \times 120 + 36 = 180 \text{ kg}$$

So, the required force is 180 kg wt = 180 × 9.807 = 1765.26 N.

A ladder 6 m long has a weight of 18 kgf and its center of gravity is 2.4 m from the bottom. The ladder is placed against a vertical wall so that it makes an angle of 60° with the ground. How far up the ladder can a 72-kgf man climb before the ladder is on the verge of slipping? The angle of friction at all contact surfaces is 15° .



coefficient of friction

$$\mu = \tan \phi = \tan 15^\circ$$

Ant-g friction at contact surfaces

$$f_A = \mu N_A = N_A \tan 15^\circ$$

$$f_B = \mu N_B = N_B \tan 15^\circ$$

Now

$$\Sigma F_y = 0: N_A + f_B = 18 + 72,$$

$$N_A = 90 - f_B; N_A = 90 - N_B \tan 15^\circ$$

$$\Sigma F_H = 0: f_A = N_B; N_A \tan 15^\circ = N_B$$

$$(90 - N_B \tan 15^\circ) \tan 15^\circ = N_B$$

$$90 \tan 15^\circ - N_B \tan^2 15^\circ = N_B$$

$$\therefore N_B = \frac{90 \tan 15^\circ}{1 + \tan^2 15^\circ} \text{ or } N_B = 22.5 \text{ kgf}$$

$$\Sigma M_A = 0: N_B(6 \sin 60^\circ) + f_B(6 \cos 60^\circ) = 18(2.4 \cos 60^\circ) + 72(x \cos 60^\circ)$$

$$\text{or } N_B(6 \tan 60^\circ) + 6 f_B = 18(2.4) + 72x$$

$$\therefore 6(22.5) \tan 60^\circ + 6(6.03) = 43.2 + 72x$$

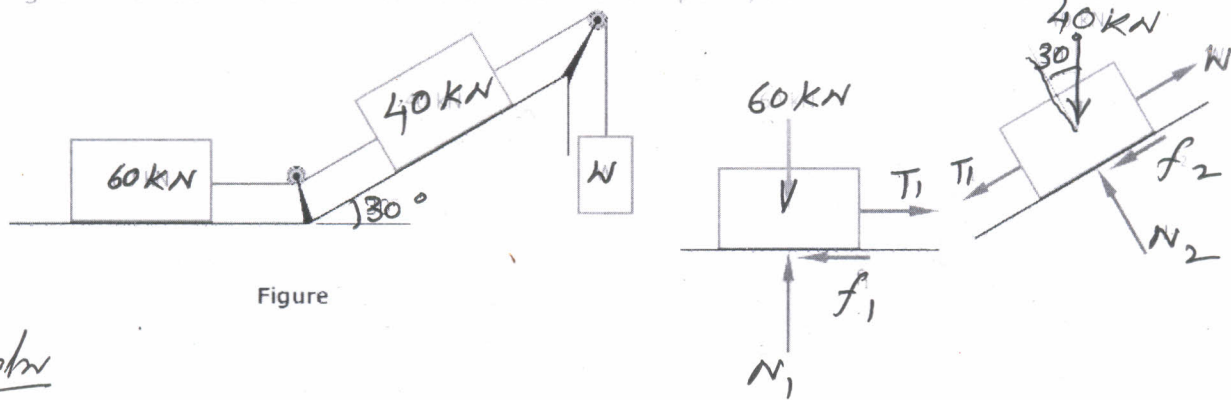
$$\therefore x = 3.15 \text{ m} \rightarrow$$

Again

$$f_B = 22.5 \text{ kgf} \tan 15^\circ = 6.03 \text{ kgf}$$

Problem

What weight W is necessary to start the system of blocks shown in Fig. moving to the right? The coefficient of friction is 0.10 and the pulleys are assumed to be frictionless.



Soln

$$N_1 = 60 \text{ kN}; f_1 = \mu N_1 = 0.10(60) = 6 \text{ kN}; T_1 = f_1 = 6 \text{ kN}$$

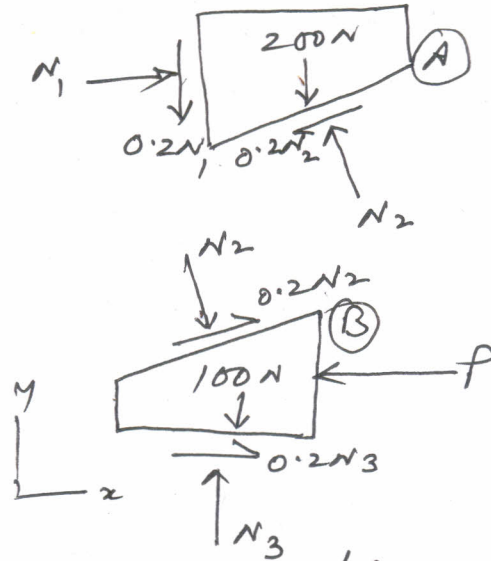
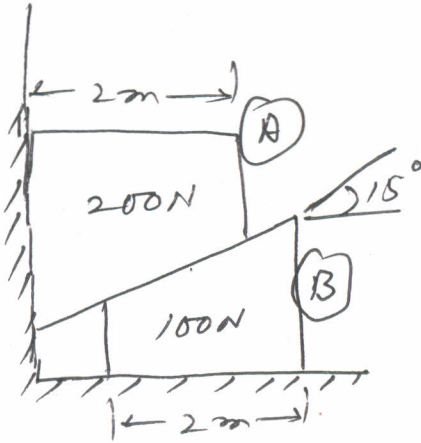
$$N_2 = 40 \cos 30^\circ = 34.64 \text{ kN}; f_2 = \mu N_2 = 0.10(34.64) = 3.46 \text{ kN}$$

$$W = 40 \sin 30^\circ + T_1 + f_2$$

$$= 20 + 6 + 3.46 = 29.46 \text{ kN (Ans)}$$

—X—

The coefficient of static friction for all surfaces in fig. is 0.2. Find out the min force 'P' required to push ^(impending) Block B to the left. Also find the 'P' required for impending motion of Block B to right.



Impending motion of B to the left
[FBD]

For FBD of 'A'

$$\Sigma F_x = 0: -N_2 \sin 15^\circ + N_1 - 0.2 N_2 \cos 15^\circ = 0 \quad \text{--- (i)}$$

$$\Sigma F_y = 0: N_2 \cos 15^\circ - 0.2 N_1 - 200 - 0.2 N_2 \sin 15^\circ = 0 \quad \text{--- (ii)}$$

Solving (i) & (ii) $N_2 = 243 \text{ N}, N_1 = 109.8 \text{ N}$

For FBD of 'B'

$$\Sigma F_x = 0: N_2 \sin 15^\circ - P + 0.2 N_3 + 0.2 N_2 \cos 15^\circ = 0 \quad \text{--- (iii)}$$

$$\Sigma F_y = 0: -N_2 \cos 15^\circ + N_3 - 100 + 0.2 N_2 \sin 15^\circ = 0 \quad \text{--- (iv)}$$

Solving (iii) & (iv) $P = 174 \text{ N} \rightarrow$

For Reverse direction of P ($\rightarrow$) all the frictional forces are reversed,

For Block A

$$\Sigma F_x = 0: -N_2 \sin 15^\circ + N_1 + 0.2 N_2 \cos 15^\circ = 0 \quad \text{--- (v)}$$

$$\Sigma F_y = 0: N_2 \cos 15^\circ + 0.2 N_1 - 200 + 0.2 N_2 \sin 15^\circ = 0 \quad \text{--- (vi)}$$

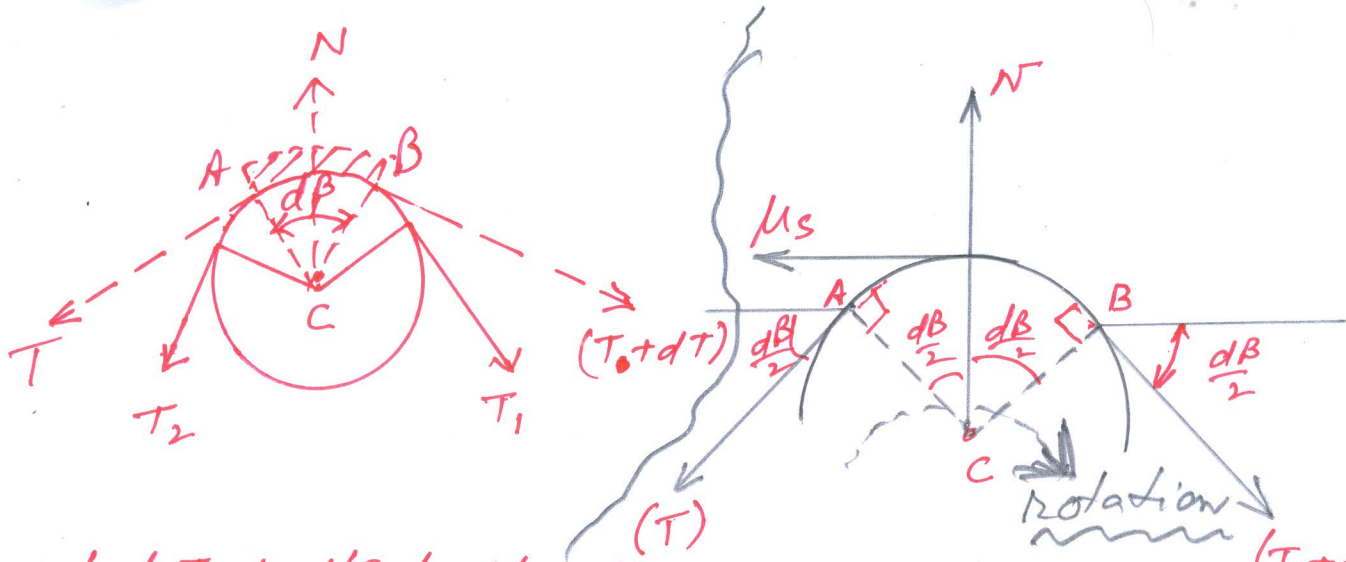
$N_2 = 194.1 \text{ N}$
 $N_1 = 12.80 \text{ N}$

For Block B

$$\Sigma F_x = 0: N_2 \sin 15^\circ + P - 0.2 N_3 - 0.2 N_2 \cos 15^\circ = 0 \quad \text{--- (vii)}$$

$$\Sigma F_y = 0: -N_2 \cos 15^\circ + N_3 - 100 - 0.2 N_2 \sin 15^\circ = 0 \quad \text{--- (viii)}$$

$P = 46.7 \text{ N} \rightarrow$



Let T_1 be the tight side, T_2 be the slack side, β = wrap angle. Consider a differential length AB of belt, for which differential wrapping angle is $d\beta$. Tight side tension is $(T + dT)$, slack side tension is T for the belt segment AB .

For no slip of belt, resistive force due to kinetic friction is μN , while the normal reaction is N .

Now $\sum F_y = 0$: $N = (T + dT) \sin \frac{d\beta}{2} + T \sin \frac{d\beta}{2}$
 as $d\beta$ is very small, $\sin \frac{d\beta}{2} \approx \frac{d\beta}{2}$ & $\cos \frac{d\beta}{2} \approx 1$.

$$\therefore \underline{N} = (T + dT) \frac{d\beta}{2} + T \frac{d\beta}{2} = \underline{T d\beta} \text{ (Neglecting } dT d\beta)$$

Again: $\sum F_x = 0$: $\mu N = (T + dT) \cos \frac{d\beta}{2} - T \cos \frac{d\beta}{2}$
 $\alpha \mu N = (T + dT) - T = dT$ — (1)

$$\text{as } \underline{N = T d\beta}$$

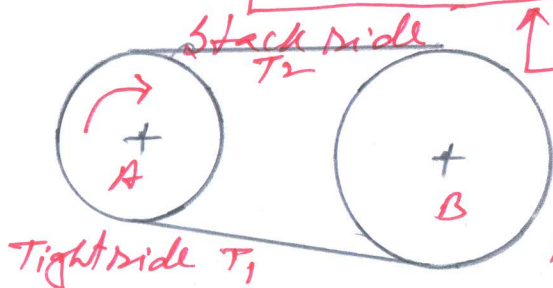
So from (1) $\mu T d\beta = dT \propto \frac{dT}{T} = \mu d\beta$

$$\alpha \int_{T_2}^{T_1} \frac{dT}{T} = \mu \int_0^\beta d\beta \propto \ln \frac{T_1}{T_2} = \mu \beta$$

$$\therefore \boxed{\frac{T_1}{T_2} = e^{\mu \beta}}$$

β in Radians

Also known as EYTELWEIN'S formula.

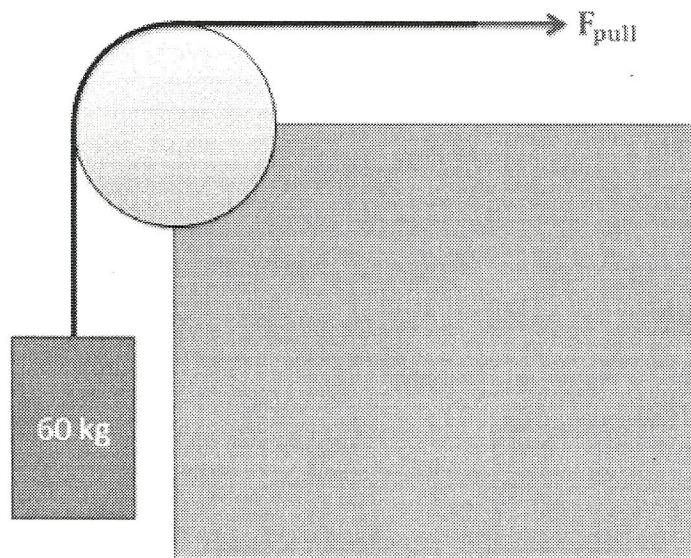


A → driver, B → driving

Question 1:

A steel cable supports a 60 kg mass and is then run a quarter of the way around a steel cylinder and supported by a pulling force as shown in the diagram below. The static coefficient of friction between the cable and the steel cylinder is 0.3.

- a • What is the minimum pulling force required to lift the mass?
- b • What is the minimum pulling force required to keep the mass from falling?



a

The free body diagram shows a circle representing the cable on the cylinder. A horizontal arrow labeled F_{pull} points to the right from the top of the circle. A vertical arrow labeled $F = (9.81)(60)$ points downwards from the left of the circle. A curved arrow labeled F_f indicates friction at the point of contact. A dashed line shows the angle of contact is $\frac{\pi}{2}$.

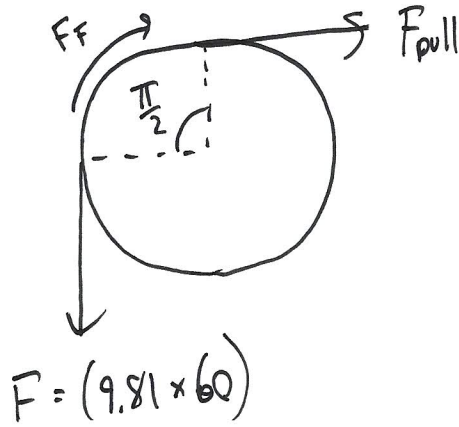
$$T_{2max} = T_1 e^{\mu_s \beta}$$

$$F_{pull} = (9.81 \times 60) N e^{(0.3)(\frac{\pi}{2})}$$

$$F_{pull} = 942.9 \text{ N}$$

to lift

b)



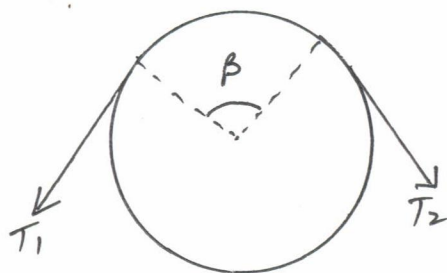
$$T_{2max} = T_1 e^{\mu_s \beta}$$

$$(9.81 \times 60) = F_{pull} e^{(.3)(\frac{\pi}{2})}$$

$$F_{pull} = 367.4 \text{ N}$$

to keep from falling

Belt Friction



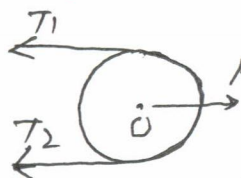
$$\frac{T_1}{T_2} = e^{\mu \beta}$$

$$T_1 > T_2$$

β in radians

The proof of this is to be done by the students as home work.

8. A drum shown in Fig below requires a torque of $200 \text{ N}\cdot\text{m}$ to get it to start rotating. If the static coefficient of friction μ_s between the belt & drum is 0.35 , what is the minimum axial force on the drum required to create enough tension in the belt to start the rotation of drum?



Torque required = $200 \text{ N}\cdot\text{m}$
Radius = 250 mm

The angle of wrap $\beta = \pi \text{ radian } (180^\circ)$.

Using the condition of impending slipping between belt & drum is OK.

$$\frac{T_1}{T_2} = e^{\mu \beta} = e^{0.35\pi} = 3.00 \quad \text{--- (1)}$$

For condition of starting rotation

$$\sum M_o = \text{Torque required} = 200 \text{ N}\cdot\text{m}$$

$$\therefore (T_1 - T_2)(0.25) = 200 \quad \text{--- (11)}$$

$$\text{From (1) \& (11)} \quad T_1 = 1200 \text{ N} \quad \& \quad T_2 = 400 \text{ N}$$

Now to get min value F ,

$$\sum F_x = 0: \quad F_{\min} = 1200 + 400 = 1600 \text{ N} \quad \text{Ans}$$

— a —

Belt Friction

In Fig., a rotating drum is braked by a flat belt attached to the lever ABC hinged at B. Determine the braking moment exerted by a vertical weight of 150N, while the μ between belt & drum is 0.43.

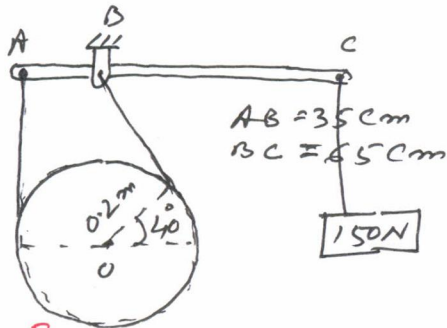
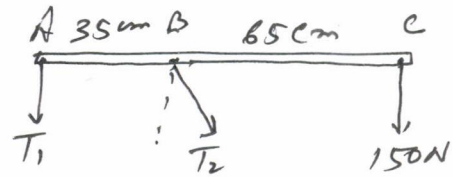


Fig.



FBD of drum



FBD of lever

The wrap angle $\beta = 180^\circ + 40^\circ = 220^\circ = 1.222\pi$ radians.

From FBD of drum, $\frac{T_1}{T_2} = e^{\mu\beta}$ or $T_1 = T_2 e^{(0.43 \times 1.222\pi)} = 5.21 T_2$ (1)

From FBD of lever ABC; Taking moment about B: $T_1 \times 0.35 = 150 \times 0.65$

$$\therefore T_1 = 278.5 \text{ N}$$

Putting value of T_1 in (1) we get $T_2 = 53.47 \text{ N}$

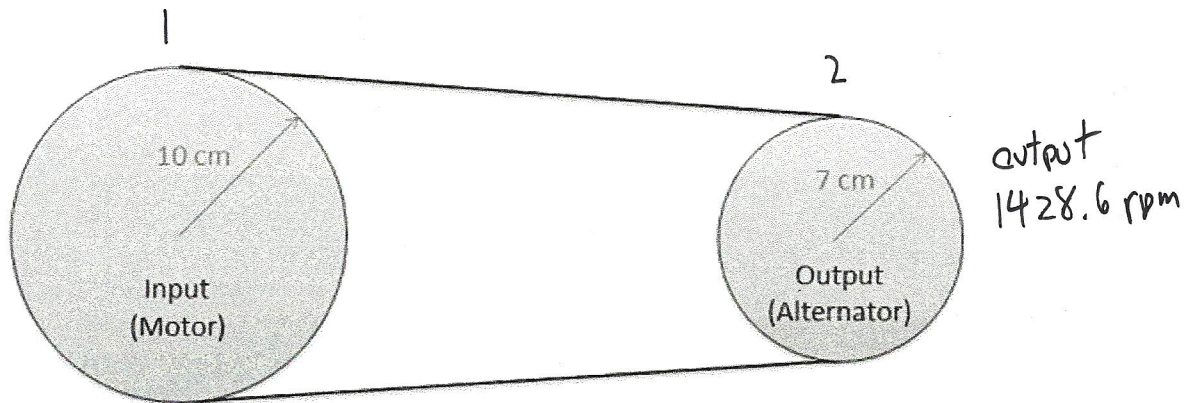
Therefore braking moment acting on the drum:

$$M = (T_1 - T_2) \times R$$

$$= (278.57 - 53.47) \times 0.2 = \underline{45.02 \text{ N-m}}$$

Question 300

A flat belt is being used to transfer power from a motor to an alternator as shown in the diagram below. The coefficient of friction between the belt material and the pulley is .5. If we require a power of 100 Watts (Nm/s) while the input is rotating at a rate of 1000 rpm, what is the required resting tension in the belt? (Assume contact angles of approximately 180°)



$$P_1 = P_2 = 100 \text{ Nm/s}$$

$$\omega_1 = 1000 \text{ rpm} = 104.7 \text{ rad/s}$$

$$\omega_2 = 1428.6 \text{ rpm} = 149.6 \text{ rad/s}$$

$$P = M \omega$$

$$100 \frac{\text{Nm}}{\text{s}} = (M)(104.7 \text{ rad/s})$$

$$M = .955 \text{ Nm}$$

$$M = (T_{2\text{max}} - T_1)(r_1)$$

$\uparrow$ $\uparrow$
 T_{rest} $.1 \text{ m}$

$$.955 \text{ N} = T_{2\text{max}} - T_{\text{rest}}$$

input
pulley

input
pulley

$$T_{2\max} = T_{\text{rest}} e^{\mu_s \beta}$$

$$T_{2\max} = T_{\text{rest}} e^{(.5)(\pi)}$$

$$T_{2\max} = 4.81 T_{\text{rest}}$$

$$9.55 \text{ N} = 4.81 T_{\text{rest}} - T_{\text{rest}}$$

$$\boxed{T_{\text{rest}} = 2.51 \text{ N}}$$

$$P = M \omega$$

$$100 \frac{\text{Nm}}{\text{s}} = (M)(149.6 \text{ rad/s})$$

$$M = .668 \text{ Nm}$$

$$M = (T_{2\max} - T_{\text{rest}}) \left(\underset{\substack{\uparrow \\ .07 \text{ m}}}{r_2} \right)$$

$$9.55 \text{ N} = T_{2\max} - T_{\text{rest}}$$

$$T_{2\max} = T_{\text{rest}} e^{\mu_s \beta}$$

$$(.5)(\pi)$$

$$T_{2\max} = T_{\text{rest}} e$$

$$T_{2\max} = 4.81 T_{\text{rest}}$$

$$9.55 \text{ N} = 4.81 T_{\text{rest}} - T_{\text{rest}}$$

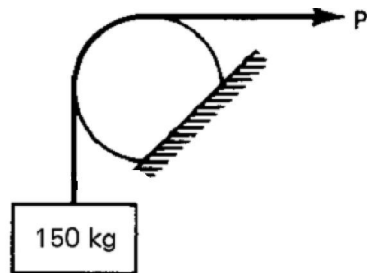
$$\boxed{T_{\text{rest}} = 2.51 \text{ N}}$$

same in both locations

output
pulley

Model questions for practice - Belt Friction

Q. The force P on the belt is just enough to raise the block. The coefficient of friction between belt and solid surface is $1/3$.



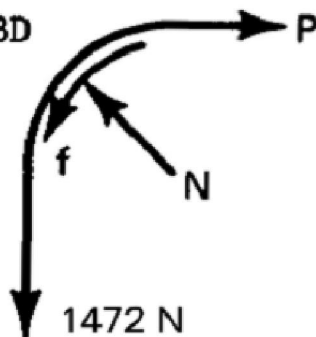
Solution:

$$\beta = \frac{\pi}{2}$$

$$T_L = P$$

$$T_S = 1472 \text{ N}$$

from the FBD



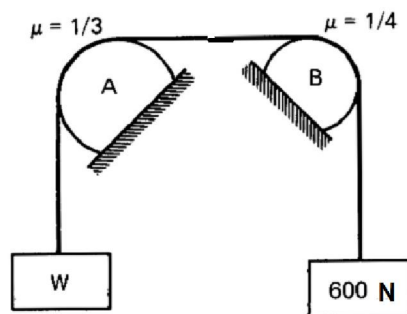
$$\frac{T_L}{T_S} = e^{\mu\beta}$$

$$\frac{P}{1472} = e^{1/3(\frac{\pi}{2})}$$

$$P = 1472 e^{\pi/6}$$

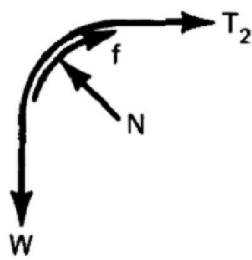
$$P = 2480 \text{ N}$$

Q. The unknown weight W is about to move downward.



Solution:

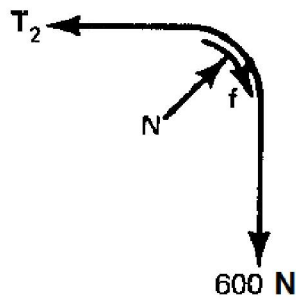
On cylinder A,



$$T_{L_A} = W \quad \text{and} \quad T_{S_A} = T_2$$

$$\text{so } \frac{T_L}{T_S} = \frac{W}{T_2} = e^{\mu\beta} = e^{\left[\frac{1}{3}\right]\left[\frac{\pi}{2}\right]} = e^{\frac{\pi}{6}}$$

On cylinder B



$$T_{L_B} = T_2$$

$$T_{S_B} = 600$$

$$\frac{T_2}{600} = e^{\frac{\pi}{8}}$$

$$\frac{W}{T_2} = e^{\frac{\pi}{6}} \quad \text{on A,} \quad \frac{T_2}{600} = e^{\frac{\pi}{8}} \quad \text{and on B.}$$

$$\frac{W}{T_2} \left[\frac{T_2}{600} \right] = e^{\frac{\pi}{6}} \left(e^{\frac{\pi}{8}} \right)$$

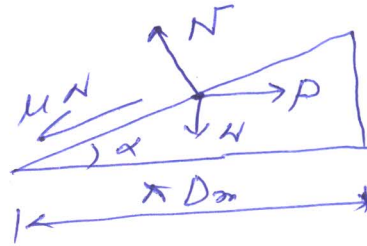
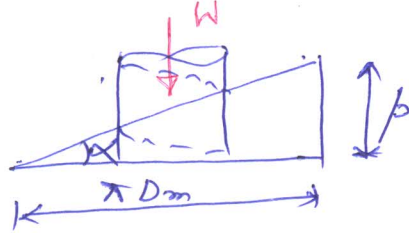
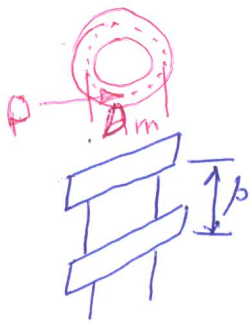
$$\frac{W}{600} = e^{\pi \left[\frac{1}{6} + \frac{1}{8} \right]}$$

$$W = 600e^{7\pi/24} = 1500$$

$$W = 1500 \text{ N}$$

Screw Jack

Page - 1



$$\sum F_x = 0: P = \mu N \cos \alpha + N \sin \alpha \quad \text{--- (1)}$$

$$\sum F_y = 0: W + \mu N \sin \alpha = N \cos \alpha$$

$$\therefore W = N \cos \alpha - \mu N \sin \alpha \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2} \quad \frac{P}{W} = \frac{\mu \cancel{N} \cos \alpha + \cancel{N} \sin \alpha}{\cancel{N} \cos \alpha - \mu \cancel{N} \sin \alpha} = \frac{\mu \cos \alpha + \sin \alpha}{\cos \alpha - \mu \sin \alpha}$$

$$\therefore \frac{P}{W} = \frac{\mu + \tan \alpha}{1 - \mu \tan \alpha} = \frac{\tan \phi + \tan \alpha}{1 - \tan \phi \tan \alpha} = \tan(\phi + \alpha)$$

$$\therefore \boxed{P = W \tan(\phi + \alpha)} \quad \text{--- (A)}$$

The effort P required to lower the load can be found

$$\text{as } \boxed{P = W \tan(\phi - \alpha)} \quad \text{--- (B)}$$

The moment required to raise the load W can be written

$$\text{as } \boxed{M = P \times \frac{D_m}{2} = \frac{W D_m}{2} \tan(\phi + \alpha) \text{ N-m}} \quad \text{--- (C)}$$

The moment required to lower the load W can be written

$$\text{as } \boxed{M = \frac{W D_m}{2} \tan(\phi - \alpha) \text{ N-m}} \quad \text{--- (d)}$$

When $\phi > \alpha$ eqⁿ (d) is (+)ve value

When $\phi < \alpha$ eqⁿ (d) is (-)ve value

When $\phi = \alpha$ eqⁿ (d) is zero value

for $\phi < \alpha$, (-)ve moment will act in the screw, hence the load W will descend down without any effort, which is the case of overhauling. So some amount of effort, hence moment will be required to stop the load to descend.

Hence $P = W \tan(\alpha - \phi)$ & $M = \frac{W D_m}{2} \tan(\alpha - \phi) \text{ N-m}$ when $\phi < \alpha$.

When $\phi = \alpha$

Page-2

Moment is zero. So the load will not descend down in the case $\phi = \alpha$.

When $\phi > \alpha$, (+)ive effort will be required to descend the load. This condition is known as SELF LOCKING CONDITION. This is desired for safe operation.

Efficiency of a square threaded screw used for screw Jack to raise the load.

$$\eta = \frac{\text{Output}}{\text{Input}} = \frac{W \cdot p}{P \cdot \pi D_m} = \frac{W \tan \alpha}{P} = \frac{W \tan \alpha}{W \tan(\phi + \alpha)} = \frac{\tan \alpha}{\tan(\phi + \alpha)}$$

$$\therefore \eta = \frac{\sin \alpha}{\cos \alpha} \times \frac{\cos(\phi + \alpha)}{\sin(\phi + \alpha)} = \frac{2 \sin \alpha \cos(\phi + \alpha)}{2 \cos \alpha \sin(\phi + \alpha)} = \frac{\sin(2\alpha + \phi) - \sin \phi}{\sin(2\alpha + \phi) + \sin \phi}$$

For η to be maximum $\sin(2\alpha + \phi)$ should be 1

$$\therefore \eta = \frac{1 - \sin \phi}{1 + \sin \phi} \quad \& \quad \sin(2\alpha + \phi) = 1 \quad \& \quad \alpha = 45^\circ - \frac{\phi}{2}$$

As for safe operation (Self locking) $\phi > \alpha$. Hence, the η in this case can be found from when

$$\eta = \frac{\tan \alpha}{\tan(\phi + \alpha)} \quad \rightarrow \text{Limiting the value of } \alpha \text{ as } \phi$$

$$\eta \leq \frac{\tan \phi}{\tan(\phi + \phi)} = \frac{\tan \phi}{\tan 2\phi} = \frac{\tan \phi (1 - \tan^2 \phi)}{2 \tan \phi} \rightarrow \text{for the case when } \alpha = \phi$$

$$\therefore \eta \leq \left[\frac{1}{2} - \frac{\tan^2 \phi}{2} \right]. \text{ This relation shows that the}$$

η of the screw to raise the load can not exceed 50% for SELF LOCKING CONDITION.

Screw Jack

Model questions on screw jack for practice

- ① Define pitch, lead, helix angle, angle of friction.
- ② Derive the effort required to raise a load of W for a pitch = l , α = helix angle & PCD of screw D_m in the form $P = W \tan(\phi + \alpha)$
- ③ Derive the effort required to lower a load of W for a pitch = l , α = helix angle, PCD of screw D_m in the form $P = W \tan(\phi - \alpha)$
- ④ What is self locking condition of a screw jack.
- ⑤ Derive the expression for η of a screw jack.
- ⑥ Derive the expression for $\max \eta$ of a screw jack.
- ⑦ Derive the expression for η of a screw jack considering self locking condition.
- ⑧ Show that the η of a screw jack under self locking condition is always less than 50%.
- ⑨ Numerical problems to find effort req to raise & lower the load. Moment req to raise & lower the load.
- ⑩ Numerical problems to find $\min$ effort required to stop the load to descend down (when self locking condition is not present). [use $P = W \tan(\alpha - \phi)$, as $\phi < \alpha$]
- ⑪ Calculate the η of a screw jack under ^{non} self locking condition & $\max \eta$. (Numerical problems)
- ⑫ Calculate the η of a screw jack under self locking condition. (Numerical problems)

MODEL QUESTIONS ON SCREW JACK FOR PRACTICE

1.

A load of 4000 N is to be lifted by a screw-jack, having threads of 10 mm pitch. The efficiency of the jack at this load is 40%. Determine the effort applied at the end of a handle of 60 cm length. [Ans. 26.5 N]

2.

Find the effort required to apply at the end of a handle, fitted to the screw hand of a screw-jack to lift a load of 2000 N. The length of the handle is 50 cm. The mean diameter and the pitch of the screw-jack are 6 cm and 10 mm respectively. The co-efficient of friction is given as 0.09. [Ans. 17.245 N]

(Hint: Force and effort are different. Effort is like work, here it is equivalent to Torque or moment(N-m). Whereas force is in Newton only. Hence EFFORT required to raise the load at mean radius is the same as Effort required at the end of any length of handle. Effort is a constant quantity, it does not change with handle or without handle. FORCE required to raise the load will depends on at what radius we are applying the force. Force required to raise the load will be less with using handle.)..... So dont get confused!. Hence here " the handle is 50 cm" has nothing to do with the calculation. But it is given to judge the understanding of the students.

3. In the above question 3,

instead of raising the load of 2000 N, the same load is lowered, determine the effort required to apply at the end of the handle.

4.

A screw-jack is used to lift a load of 4 kN. The screw of the screw-jack is square threaded with two threads to 1 cm. If the co-efficient of friction between the nut and screw is 0.095 and outer diameter of screw is 6 cm, find the force required at the end of the handle of length 50 cm to lift the load. [Ans. 28.29 N]

5.

A screw-jack has a square thread, 6 cm mean diameter and 1 cm pitch. The load on the jack revolves with the screw. The co-efficient of friction at the screw threads is 0.045 :

(i) Find the tangential force to be applied to the jack at 40 cm radius so as to lift a load of 700 N.

(ii) State whether the jack is self-locking. If it is, find the torque necessary to lower the load. If not, find the torque, which must be applied to keep the load from descending. [Ans. (i) 5.16 N, (ii) No, .1675 Nm]

6.

A screw-jack has a square thread of mean diameter 6 cm and pitch 9 mm. The co-efficient of friction at the screw-thread is 0.09. A load of 2 N is to be lifted through 10 cm. Determine the torque required and work done in lifting the load through 10 cm. Find the efficiency of the jack also. [Ans. .00829 kN ; 5792 kN ; 34.53%]